

lab04

AUTHOR

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1.Carregue os pacotes readxl e tidyverse.

```
library(readxl)
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    3.5.2    ✓ tibble     3.2.1
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.0.4
```

— Conflicts — tidyverse_conflicts() —

```
✗ dplyr::filter() masks stats::filter()
```

```
✗ dplyr::lag() masks stats::lag()
```

! Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become errors

2.Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists

```
fname <- "WDIEXCEL.xlsx"

file.exists(fname)
```

```
[1] TRUE
```

3.Leia a planilha "Country" e armazene no objeto countries.

```
countries <- read_excel(fname, sheet = "Country")

countries
```

```
# A tibble: 264 × 31
```

	`Country Code`	`Short Name`	`Table Name`	`Long Name`	`2-alpha code`
	<chr>	<chr>	<chr>	<chr>	<chr>
1	ABW	Aruba	Aruba	Aruba	AW
2	AFE	Africa Eastern and So...	Africa East...	Africa Eas...	ZH
3	AFG	Afghanistan	Afghanistan	Islamic St...	AF
4	AFW	Africa Western and Ce...	Africa West...	Africa Wes...	ZI
5	AGO	Angola	Angola	People's R...	AO
6	ALB	Albania	Albania	Republic o...	AL
7	AND	Andorra	Andorra	Principali...	AD
8	ARB	Arab World	Arab World	Arab World	1A
9	ARE	United Arab Emirates	United Arab...	United Ara...	AE
10	ARG	Argentina	Argentina	Argentine ...	AR

```
# i 254 more rows
```

```
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries <- countries %>%
  filter(`Country Code` %in% c("BRA", "ARG", "CHL", "MEX", "USA"))

countries
```

```
# A tibble: 5 × 31
  `Country Code` `Short Name` `Table Name` `Long Name`      `2-alpha code`
  <chr>          <chr>        <chr>      <chr>          <chr>
1 ARG           Argentina  Argentina  Argentine Republic AR
2 BRA           Brazil     Brazil     Federative Republic... BR
3 CHL           Chile      Chile      Republic of Chile    CL
4 MEX           Mexico     Mexico     United Mexican Stat... MX
5 USA           United States United States United States of Am... US
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lg1>,
#   `PPP survey year` <lg1>, `Balance of Payments Manual in use` <chr>, ...
```

5. Leia a planilha "Data" e armazene no objeto WDI.

```
WDI <- read_excel(
  fname,
  sheet = "Data"
)
```

6. Selecione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)

WDI
```

```
# A tibble: 396,970 × 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>        <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 AFE           EG.CFT.ACCS.ZS      NA     NA     NA     NA     NA     NA
```

```

2 AFE      EG.CFT.ACCS.RU.ZS      NA      NA      NA      NA      NA      NA
3 AFE      EG.CFT.ACCS.UR.ZS      NA      NA      NA      NA      NA      NA
4 AFE      EG.ELC.ACCS.ZS          NA      NA      NA      NA      NA      NA
5 AFE      EG.ELC.ACCS.RU.ZS      NA      NA      NA      NA      NA      NA
6 AFE      EG.ELC.ACCS.UR.ZS      NA      NA      NA      NA      NA      NA
7 AFE      FX.OWN.TOTL.ZS          NA      NA      NA      NA      NA      NA
8 AFE      FX.OWN.TOTL.FE.ZS      NA      NA      NA      NA      NA      NA
9 AFE      FX.OWN.TOTL.MA.ZS      NA      NA      NA      NA      NA      NA
10 AFE     FX.OWN.TOTL.OL.ZS      NA      NA      NA      NA      NA      NA

```

```
# i 396,960 more rows
```

```

# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...

```

7.Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```

WDI <- WDI %>%
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "VALUE"
  ) %>%
  mutate(YEAR = as.integer(YEAR))

```

```
WDI
```

```
# A tibble: 23,421,230 × 4
```

```

`Country Code` `Indicator Code` YEAR VALUE
<chr>          <chr>          <int> <dbl>
1 AFE          EG.CFT.ACCS.ZS      1960    NA
2 AFE          EG.CFT.ACCS.ZS      1961    NA
3 AFE          EG.CFT.ACCS.ZS      1962    NA
4 AFE          EG.CFT.ACCS.ZS      1963    NA
5 AFE          EG.CFT.ACCS.ZS      1964    NA
6 AFE          EG.CFT.ACCS.ZS      1965    NA
7 AFE          EG.CFT.ACCS.ZS      1966    NA
8 AFE          EG.CFT.ACCS.ZS      1967    NA
9 AFE          EG.CFT.ACCS.ZS      1968    NA
10 AFE         EG.CFT.ACCS.ZS      1969    NA

```

```
# i 23,421,220 more rows
```

8.Confirme se a coluna referente ao ano está em formato inteiro.

```
is.integer(WDI$YEAR)
```

```
[1] TRUE
```

9.Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```
WDI <- WDI %>%
  filter(
    `Country Code` %in% countries$`Country Code`,
    `Indicator Code` == "SP.DYN.CBRT.IN"
  ) %>%
  left_join(
    countries %>% select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  select(`Country Code`, `Short Name`, YEAR, VALUE) %>%
  rename(`SP.DYN.CBRT.IN` = VALUE)
```

WDI

```
# A tibble: 295 × 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 ARG           Argentina  1960      24.2
2 ARG           Argentina  1961      23.9
3 ARG           Argentina  1962      23.8
4 ARG           Argentina  1963      23.5
5 ARG           Argentina  1964      23.2
6 ARG           Argentina  1965      22.6
7 ARG           Argentina  1966      22.3
8 ARG           Argentina  1967      22.4
9 ARG           Argentina  1968      22.7
10 ARG          Argentina  1969      23.0
# i 285 more rows
```

10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```
WDI <- WDI %>%
  arrange(`Country Code`, YEAR)
```

WDI

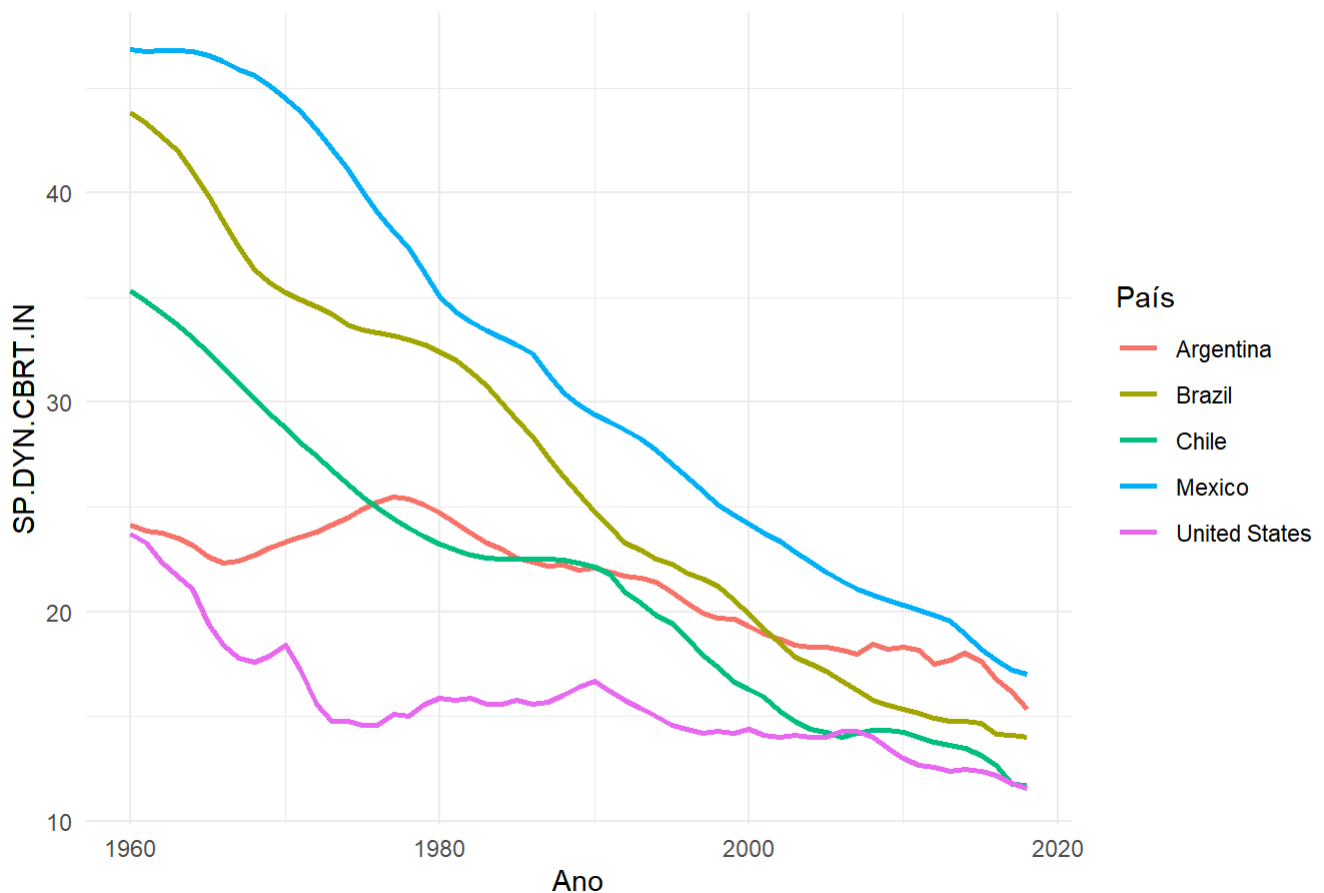
```
# A tibble: 295 × 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 ARG           Argentina  1960      24.2
2 ARG           Argentina  1961      23.9
3 ARG           Argentina  1962      23.8
4 ARG           Argentina  1963      23.5
5 ARG           Argentina  1964      23.2
6 ARG           Argentina  1965      22.6
7 ARG           Argentina  1966      22.3
8 ARG           Argentina  1967      22.4
9 ARG           Argentina  1968      22.7
```

```
10 ARG      Argentina      1969      23.0
# i 285 more rows
```

11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
ggplot(
  WDI,
  aes(
    x = YEAR,
    y = `SP.DYN.CBRT.IN`,
    color = `Short Name`,
    group = `Short Name`
  )
) +
  geom_line(na.rm = TRUE, linewidth = 1) +
  labs(
    title = "Taxa bruta de natalidade – 1960 a 2018",
    x = "Ano",
    y = "SP.DYN.CBRT.IN",
    color = "País"
  ) +
  theme_minimal()
```

Taxa bruta de natalidade — 1960 a 2018



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 21:12:33 -03"
```